

READING PASSAGE 1

The Physics of Tornadoes and Severe Weather

A The study of the physics of tornadoes and severe weather has undergone a remarkable transformation over the past several decades. What was once a niche area of academic inquiry has developed into a rich and diverse field of research, drawing on insights from a wide range of disciplines and theoretical traditions.

B The intellectual lineage of this field extends considerably further than many realise. While modern research techniques have only recently been applied to these questions, philosophers and natural scientists have grappled with related issues for centuries. The transition from speculative philosophical inquiry to systematic empirical investigation represents one of the most significant developments in the intellectual history of this domain.

C At the core of current understanding lies a set of interrelated mechanisms that collectively explain the observed patterns and outcomes. First, structural factors create the underlying conditions within which individual and collective action takes place. These structural dimensions include economic incentives, institutional frameworks, and cultural norms that shape behaviour in ways that are often not immediately apparent to the actors themselves. Second, dynamic feedback processes amplify or dampen initial effects, generating patterns of change that can diverge substantially from what linear models would predict.

D Case studies from diverse geographical and cultural settings provide compelling illustrations of these dynamics in action. In one particularly well-documented example, researchers tracked developments over a fifteen-year period in a setting that offered a natural experiment in the effects of policy change. The introduction of a new regulatory framework produced a cascade of adaptations that rippled through the entire system, generating outcomes that both confirmed theoretical predictions and revealed previously unrecognised complexities.

E Despite the substantial progress that has been made, significant challenges and controversies remain. Critics have raised important methodological concerns regarding the generalisability of findings derived from studies conducted in specific contexts. The predominance of research carried out in Western, educated, industrialised, rich, and democratic societies raises questions about the applicability of current theories to populations with different cultural, economic, and political circumstances. Furthermore, the reliance on cross-sectional designs in much of the existing literature limits the capacity to draw strong causal inferences.

F The implications of this body of research extend well beyond the academy. As societies grapple with increasingly complex challenges, the insights generated by scholarly inquiry in this domain have the potential to inform more effective and equitable policy responses. Realising this potential, however, will require sustained investment in both fundamental research and the translational infrastructure needed to convert knowledge into action.

G The evidence reviewed above demonstrates both the substantial achievements of research in this field and the significant work that remains to be done. As new tools and theoretical perspectives become available, and as the phenomena themselves continue to evolve, the research community will need to maintain both rigour in its methods and openness to new ideas. The stakes are high, for the questions addressed by this research bear directly on some of the most important challenges facing humanity in the twenty-first century.

Question 1

Choose the correct letter, A, B, C or D.

- 1 The author suggests that early studies in this field were limited by:
- A. inconsistent results across cultures
 - B. a consistent pattern across diverse contexts
 - C. no measurable effects
 - D. effects limited to Western countries

Question 2

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 2 The author notes that researchers have called for more _____ approaches to investigation.

Question 3

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 3 Early researchers had access to the same technologies as modern scholars.

Questions 4–5

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 4 The landmark study analysed data spanning a period of _____ years.
- 5 Case studies have documented the effects of _____ change on the system.

Question 6

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 6 The passage suggests that structural factors are less important than individual choices.

Question 7

Choose the correct letter, A, B, C or D.

- 7 What finding emerged from the landmark study discussed in the passage?
- A. economic incentives and institutional constraints
 - B. individual personality traits
 - C. random chance
 - D. government policies alone

Questions 8–9

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 8 Meta-analyses have confirmed the reliability of the core findings in this field.
- 9 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.

Question 10

Choose the correct letter, A, B, C or D.

- 10 Which of the following best describes the author's view on future developments?
- A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations

Question 11

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 11 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Questions 12–13

Choose the correct letter, A, B, C or D.

- 12 According to the text, what role do structural factors play?
- A. pessimism about the future
 - B. cautious optimism with acknowledgment of challenges
 - C. complete certainty about outcomes
 - D. indifference to practical applications
- 13 What does the passage identify as a significant challenge to current understanding?
- A. inconsistent results across cultures
 - B. a consistent pattern across diverse contexts
 - C. no measurable effects
 - D. effects limited to Western countries